

Lecture 3: Chapters 6, 18- Thermodynamics and chemical reactions

Concept	Explanation	
Isolated System	Neither matter or heat can be transported across the bord (internal energy is conserved), completely self contained	
Closed System	Matter cannot be transported across the border – but heat can. Can gain or lose energy, but its mass remains constant	
Spontaneous change	- Occurs without a continuous input of energy from outside the system. - Moves a system towards equilibrium - Increase in entropy - All chemical processes require energy (activation energy) to take place, but once a spontaneous process has begun, no further input of energy is needed. - A spontaneous process in one direction can not be executed in the other under the same conditions (it would be non spontaneous in the other direction) (ex. Melting ice into water)	
Non Spontaneous change	Occurs only if the surroundings continuously supply energy to the system. If a change is spontaneous in one direction, the reverse direction is nonspontaneous Move the system away from equilibrium, decreasing the entropy (ex. Freezing water)	
Entropy (S)	Measure of how spread out or dispersed the energy of a system is among the different possible ways the system can contain energy More dispersed = more entropy Maximum at equilibrium (forward reverse properties at equal rate)	
States of Matter	Phases: Solid (s): Fills part of the volume, retains a permanent form. Liquid (l): Fills part of the volume, fluid in behavior. Gas (g): Fills the entire volume, fluid in nature	Phase Transitions: Condensation: Gas → Liquid Evaporation: Liquid → Gas Freezing: Liquid → Crystalline solid Melting: Crystalline solid → Liquid Sublimation: Crystalline solid → Gas Deposition: Gas → Crystalline solid

Atmospheric Composition	UV Light: UVA: 315-400 nm UVB: 280-315 nm UVC: 100-280 nm	Ozone Formation: $O_2 + h\nu \rightarrow 2O$ $O_2 + O \rightarrow O_3$
Internal Energy (U)	State function: its value depends only on the current state of the system, not on the path taken to reach that state (determined by factors like temp, pressure and volume), will be the same for any given state	Includes all Kinetic and Potential energy of a system
Transition State (Activated Complex)	- High energy unstable state (peak energy) - Old bonds are partially broken, and new bonds are partially formed - Transition state is reached by the activation energy	

Formula	Variables	Use + Additional Information	Example
$pV=nRT$ <i>Ideal Gas Law</i>	P: Pressure (Pa, atm) V: Volume (m^3 , L) n: number of moles (mol) R: Gas Constant ($0.08206 \frac{L \cdot atm}{mol \cdot K}$) T: Temperature(K)	Used to calculate the state variables (pressure, volume, temperature) of an ideal gas. Solve problems involving gas mixtures, reactions involving gaseous products, and standard conditions. Assumptions about gas molecules: - Point particles - In constant motion - Collisions are perfectly elastic - Do not have forces between them	Calculate the volume of 1 mole of an ideal gas at standard temperature and pressure (STP: p=1 atm, T=273.15 K Solution: $V=nRT/p=(1 \text{ mol})(0.08206 \text{ L} \cdot \text{atm/mol} \cdot \text{K})(273.15 \text{ K})/1 \text{ atm}$ $\approx 22.4 \text{ L}$
<i>The Mean Free Path λ</i>			

$K.E. = 0.5mv^2$ Kinetic Energy of a gas molecule	M: mass of a gas molecule (kg) V: speed of the gas molecules (m/s)	Used to describe the energy associated with the motion of gas molecules. Fundamental to understanding gas behavior in kinetic theory and diffusion processes.	Find the kinetic energy of an oxygen molecule $M = 2.66 \times 10^{-26} \text{ kg}$ $V = 500 \text{ m/s}$ $K.E. = \frac{0.5 \times 2.66 \times 10^{-26}}{500^2}$ $= 3.325 \times 10^{-21}$
$\Delta U = q + w$ 1st Law of Thermodynamics Energy can not be created or destroyed	ΔU : change in internal energy (J) q: heat added to the system (J) W: work done on the system (J)	Helps analyze energy changes in systems under various processes (e.g., isothermal, adiabatic). Useful in calorimetry and thermodynamic efficiency calculations. $q < 0$ Heat released from the system to the surroundings (exothermic process) $q > 0$ Heat absorbed by the system from the surroundings (endothermic process) $w < 0$ work done by the system on the surroundings $w > 0$ work done on the system by the surroundings	
Work Done in gas expansion or compression $w = -P\Delta V$	W: work P: Pressure V: Volume		
$\Delta G^\circ = -RT \ln(K)$ The standard Gibbs free energy change relation to equilibrium constant	ΔG_r° Standard Gibbs Free Energy of a reaction (J) R: gas constant T: Temperature (K) K: equilibrium constant	Will a reaction happen in standard conditions? How favorable? If $K > 1$, $\Delta G_r^\circ < 0$ The reaction is spontaneous under standard conditions. - Reaction favours product formation	

		If $K < 1$, $\Delta G^\circ > 0$ The reaction is non-spontaneous under standard conditions. - Reaction favours reactant formation If $K = 1$, $\Delta G^\circ = 0$ The reaction is at equilibrium.	
$\Delta G = \Delta G^\circ + RT \ln(Q)$ Relationship between free energy change and standard free energy change and reaction quotient	Q - reaction quotient	If a reaction is spontaneous at that specific point If $Q < 1$: The concentration of products is lower than concentration of reactants. Formation of reactants is favoured. So non spontaneous If $Q > 1$: The concentration of products is higher than concentration of reactants. Formation of products is favoured. So spontaneous	
Gibbs Free Energy $\Delta G = \Delta H - T\Delta S$	ΔG: Change in Gibbs Free Energy (J) ΔH: Change in enthalpy (heat content, J) T: Temperature (K) ΔS Change in entropy (disorder, J/K)	$\Delta G < 0$: Reaction is spontaneous. $\Delta G > 0$: Reaction is nonspontaneous. $\Delta G = 0$: Reaction is at equilibrium. Components: • Enthalpy (ΔH): Represents the heat released or absorbed. ○ $\Delta H < 0$: Exothermic (releases heat). ○ $\Delta H > 0$: Endothermic (absorbs heat). • Entropy (ΔS): Represents disorder or randomness. ○ $\Delta S > 0$: Increased disorder (favourable for spontaneity).	

		○ $\Delta S < 0$: Decreased disorder (unfavourable for spontaneity). Temperature Dependence: ● The $T\Delta S$ term shows how temperature impacts spontaneity: ○ At high temperatures, entropy (ΔS) plays a larger role. ○ At low temperatures, enthalpy (ΔH) dominates.	
Enthalpy Definition $H = U + PV$ Enthalpy change for constant Pressure $\Delta H = \Delta U + p\Delta V$ Enthalpy Change for constant temperature $\Delta U = \Delta H - RT\Delta n$	H: enthalpy (J) U: internal energy (J) P: pressure (atm) V: Volume (L/ m ³) R: gas constant T: Temperature Δn : change between moles of gaseous products and reactants	Constant Volume: $U = \Delta q$ Constant Pressure: $\Delta H = \Delta q$	
$\Delta H_{rxn} = H(p) - H(r)$ $\Delta H_{frm} = H(p) - H(r)$		Total enthalpy change for the reaction is the same, no matter the intermediate state. Energy states in reactions are independent of the path they take	
$\Delta S = \int_{T_1}^{T_2} \frac{C_p}{T} dT$	ΔS : change in entropy C_p : specific heat capacity at constant pressure ($\frac{J}{mol K}$) T: Temperature (K)		
Second Law of		Spontaneous process: the total entropy of a system and its surroundings always	

Thermodynamics $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0$ $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} = 0$		increases (predicts direction of natural processes) Equilibrium Process: total entropy of the system and surroundings is constant. (max entropy at equilibrium, no tendency for more spontaneous change)	
Standard energy changes of reactions $\Delta G_r^o = \Delta G_f^o(p) - \Delta G_f^o(r)$ $\Delta H_r^o = \Delta H_f^o(p) - \Delta H_f^o(r)$ $\Delta S_r^o = \Delta S_f^o(p) - \Delta S_f^o(r)$ State functions thermodynamics		Standard free energy changes of reactions: - Measure of reaction spontaneity under reaction conditions - G negative - spontaneous - G positive - non	
Specific Heat Capacity: $q = mc\Delta T$			

Lecture 5: Bonds, geometries, and hybridization (Ch9 - 10)

Lecture 6: Chapters 12, 13 - Inter-Molecular Interactions and solutions

Main Concept	Explanation
Inter-molecular interactions	Inter-molecular forces include: - Hydrogen Bonding: Strong attraction between a hydrogen atom bonded to an electronegative atom (N, O, F) and another electronegative atom. - Dipole-Dipole Interactions: Forces between polar molecules. - London Dispersion Forces: Weak attractions due to temporary dipoles in nonpolar molecules. These forces affect boiling/melting points, surface tension, and solubility.
Crystals	Crystals are highly organized solids with specific lattice structures: - Ionic Crystals: Formed by electrostatic attraction (e.g., NaCl). - Covalent Crystals: Held together by covalent bonds (e.g., diamond). - Metallic Crystals: Metal cations surrounded by delocalized electrons (e.g., iron). - Molecular Crystals: Composed of molecules held together by inter-molecular forces (e.g., ice). Characterized by defined shapes, symmetry, and melting points.
Phase separation	Phase separation occurs when components of a mixture separate into different physical states: - Driven by differences in polarity, density, or solubility. - Examples include oil and water or immiscible liquids. Applications: Purification processes, emulsions, and phase diagrams.
Solubility	Solubility describes the extent to which a solute dissolves in a solvent: - Factors: Temperature (e.g., solubility of salts increases with temperature), pressure (important for gases), and polarity. - Saturation levels: Unsaturated, saturated, and supersaturated solutions. Quantified by solubility constants and affected by the nature of solute-solvent interactions.

Concentration	Concentration measures the amount of solute in a solution: - Molarity (M): Moles of solute per liter of solution. - Molality (m): Moles of solute per kilogram of solvent. - Mass Percent: $(\text{Mass of solute} / \text{Mass of solution}) \times 100\%$. - Other units: Parts per million (ppm), normality, mole fraction. Used in stoichiometry and solution preparation.
Boiling point	The boiling point is the temperature at which the vapor pressure equals the external pressure: - Normal boiling point: Boiling point at 1 atm pressure. - Affected by altitude (lower pressure reduces boiling point) and inter-molecular forces. - Stronger inter-molecular forces lead to higher boiling points (e.g., water vs. ethanol).
Osmosis	Osmosis is the diffusion of a solvent across a semi-permeable membrane: - Solvent moves from a region of lower solute concentration to higher solute concentration. - Osmotic Pressure: The pressure required to stop osmosis. - Important in biological processes like nutrient absorption and maintaining cell turgor.
Viscosity	Viscosity is the measure of a fluid's resistance to flow: - Factors: Temperature (higher temperature reduces viscosity), molecular size, and inter-molecular forces. - Measured in poise (P) or centipoise (cP). Applications include lubrication, manufacturing, and studying liquid properties.
Vaporization	Vaporization is the transition of a liquid to a gas: - Requires energy to overcome inter-molecular forces (enthalpy of vaporization). - Occurs via evaporation (below boiling point) or boiling (at boiling point). Factors affecting rate: Temperature, surface area, and pressure.
Colligative properties	Colligative properties depend only on the number of solute particles, not their identity: - Boiling Point Elevation: Solute increases boiling point. - Freezing Point Depression: Solute decreases freezing point. - Vapor Pressure Lowering: Solute reduces solvent's vapor pressure. - Osmotic Pressure: Proportional to solute concentration. Used to determine molecular weights and in practical applications like antifreeze.

Equation	Variable and Unites	Use	Example
Solubility:			
Percent by mass volume: $\text{percent} = \frac{\text{mass of solute}}{\text{mass of solute} + \text{mass of solvent}}$			
Molarity: $\text{molarity} = \frac{\text{moles of solute}}{\text{liters of solution}}$	mol/L : molarity		
Molality: $\text{molality} = \frac{\text{moles of solute}}{\text{mass of solvent}}$	mol/mg : molality	Number of moles of solute dissolved in 1 mg of solvent	
Henry's Law: (solubility of gases) $C = k_H \cdot P$	C : concentration of gas in liquid k_H : Henry's constant in mol/K*atm P : pressure of gas over the solution	The solubility (c) of a gas in a liquid is directly proportional to the partial pressure (P) of the gas above the liquid.	The solubility of oxygen in water at 1 atm is 1.3×10^{-3} mol/L. What is the solubility at 2 atm? $c = kP \quad \text{so } k = \frac{1.3 \times 10^{-3}}{1} = 1.3 \times 10^{-3}$ $c = (1.3 \times 10^{-3}) \times 2 = 2.6 \times 10^{-3} \text{ mol/L}$
Raoult's Law $P_1 = X_1 P_1^0$	P_1 : Vapor pressure of the solution. P_1^0 : Vapor pressure of the pure solvent. X_1 : Mole fraction of the solvent.		If the mole fraction of water in a solution is 0.9 and the vapor pressure of pure water is 25 torr, find the vapor pressure of the solution. $P_1 = X_1 P_1^0 = 0.9 \times 25 = 22.5 \text{ torr}$
Clausius- Clapeyron equation:	P: vapour pressure(Pa, atm) ΔH_{vap} : enthalpy of vaporization (J/mol) R: gas constant (8.314 J/mol) (0.0802)	Calculates the vapour pressure	

$\ln P = -\frac{\Delta H_{vap}}{RT} + c$ Vapour pressure at diff temps: $\ln \frac{P_1}{P_2} = -\frac{\Delta H_{vap}}{R} \cdot \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$	T: temp (K)		
Colligative properties:			
Vapour-Pressure Lowering: $\Delta P = X_2 P_1^0$	ΔP: Decrease in vapor pressure. X_2: Mole fraction of the solute. P_1^0: Vapor pressure of the pure solvent.	Relates solution vapour pressure to solvent vapour pressure	For a solution with a mole fraction of solute (X_2) = 0.1 and pure solvent vapor pressure (P_1^0) = 100 torr: $\Delta P = 0.1 \times 100 = 10 \text{ torr}$
Boiling Point Elevation $\Delta T_b = K_b m$	ΔT_b: Elevation in boiling point. C K_b: Boiling point elevation constant. $\frac{C^* kg}{mol}$ m: Molality of the solution. $\frac{mol}{kg}$	c	If K_b for water is $0.512 \frac{C^* kg}{mol}$ and a solution has a molality of 2 mol/kg: $\Delta T_b = 0.512 \times 2 = 1.024 ^\circ C$
Freezing Point Depression $\Delta T_f = K_f m$	ΔT_f: Elevation in boiling point. C K_f: Boiling point elevation constant. $\frac{C^* kg}{mol}$ m: Molality of the solution. $\frac{mol}{kg}$		If K_f for water is $1.86 \frac{C^* kg}{mol}$ and a solution has a molality of 3 mol/kg: $\Delta T_f = 1.86 \times 3 = 5.58 ^\circ C$
Osmotic Pressure $\Pi = iMRT$ i = actual n° particles in soln after disoc./ n° of formula units initially dissolved.	π: Osmotic pressure (units: atm or Pa). i: van't Hoff factor, accounting for the dissociation of solutes (e.g., i=2i for NaCl, as it dissociates into Na⁺ and Cl⁻).		

	M: Molarity of the solution (mol/L). R: Gas constant T: Temperature (K).		
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Lecture 7: Chapters 14, 15 - Kinetics and Equilibrium

Equation	Variables and Units	Use	Example
First Order Reactions			
$[A] = [A]_0 e^{-kt}$	[A] : Concentration of product $[A]_0$: initial concentration of product k: rate constant T: time		
$\ln[A] = \ln[A]_0 - kt$			
$t_{\frac{1}{2}} = \frac{\ln(2)}{k}$			
Second Order Reactions			
$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$			
$t_{\frac{1}{2}} = \frac{1}{k[A]_0}$			
Arrhenius Equation:	k: rate constant		

$k = Ae^{-\frac{E_a}{RT}}$	A: frequency factor(how often molecules collide) E_a : activation energy R: gas constant T: absolute Temperature		
Graphical determination of E_a: $\ln k = \left(-\frac{E_a}{R}\right) \cdot \left(\frac{1}{T}\right) + \ln A$ y = m x + c Relationship of rate constants at different temperatures: $\ln\left(\frac{k_1}{k_2}\right) = \left(\frac{E_a}{R}\right) \cdot \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$	$-\frac{E_a}{RT}$: slope	Determine E_a by plotting $1/T$ on x-axis and $\ln k$ on y-axis	Arrhenius plot
Equilibrium Constant $K = \frac{[products]^A}{[reactants]^b}$		$K \ll 1$ Reactants favoured: dominate at equilibrium $K = 1$ Both sides equally favourable - appear in same amounts at equilibrium $K \gg 1$ Products favoured - dominate at equilibrium	
Solubility Constant $K_{sp} = [A^{n+}(aq)][B^{n-}(aq)]$			
Acid Ionization Constant $K_A = \frac{[H^+][A^-]}{[HA]}$			

Lecture 8: Chapters 11, 22 - Organic chemistry

Polymers		
2 Types of polymer synthesis reaction: -Addition -Condensation	Addition: -Contains double bonds (unsaturated) - e.g Polyethylene, Polypropylene -Contains two reactive functional groups -By-products (H₂O, HCl) -e.g nylon, polyester	
Degree of polymerization: $DP = \frac{\text{Molar mass of polymer (M)}}{\text{Molar mass of monomer (M}_m\text{)}}$	-the number of repeating units (monomers) in a polymer chain. It gives a measure of the polymer's molecular size. M	Example: Polyethylene made from ethene (C₂H₄): • Molar mass of ethene = 28 g/mol • Assume a polymer chain has a molar mass of 280,000 g/mol DP=Molar mass of polymer/Molar mass of monomer DP = 280 000/ 28 = 10 000 So, degree of polymerization is 10 000

Lecture 9: Chapters 16, 17: Acids and Bases

Main Concept	Detailed Explanation		
Acids and Bases	Acids and bases are substances that participate in proton exchange as described by the Brønsted-Lowry theory. - Acid: A substance that <u>donates</u> a proton (H+) to a base. Example: $HCl + H_2O \rightarrow H_3O^+ + Cl^-$. - Base: A substance that <u>accepts</u> a proton from an acid. Example: $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$. Conjugate Acids and Bases: Differ by one proton (H+) Such as NH_3 and NH_4^+		
	<table><tr><td>Amphoteric</td><td>Can REACT as both acids and bases (can be applied to other theories)</td></tr></table>	Amphoteric	Can REACT as both acids and bases (can be applied to other theories)
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	<table><tr><td>Amphiprotic</td><td> ARE both an acid and a base; can donate or accept protons. It must have both a lone pair and a hydrogen that can be released as H^+ Used exclusively in the Bronsted-Lowry theory of acid and bases. All amphiprotic species are amphoteric. $HCO_3^- + H_3O^+ \rightleftharpoons H_2CO_3 + H_2O$ $HCO_3^- + OH^- \rightleftharpoons CO_3^{2-} + H_2O$ </td></tr></table>	Amphiprotic	ARE both an acid and a base; can donate or accept protons. It must have both a lone pair and a hydrogen that can be released as H^+ Used exclusively in the Bronsted-Lowry theory of acid and bases. All amphiprotic species are amphoteric. $HCO_3^- + H_3O^+ \rightleftharpoons H_2CO_3 + H_2O$ $HCO_3^- + OH^- \rightleftharpoons CO_3^{2-} + H_2O$
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	Neutralization reactions with metal oxides : Acid + metal O → salt + water		

		Neutralization reactions with ammonia : Acid + ammonia → salt
		Reaction with metal carbonates : Acid + metal CO₃ → salt + water + CO₂ <i>Gas produces bubbles + effervescence</i>
		Reaction with metal hydrogen carbonates Acid + metal HCO₃ → salt + water + CO₂ <i>Gas produces bubbles + effervescence</i>
		Reaction with active metals Acid + metal → salt + H₂ <i>Reason for corrosive properties</i>
		* active metals are those above H on the activity series.
	The Lewis definition expands this by defining acids as electron pair acceptors and bases as electron pair donors. Applications: Understanding chemical reactivity in biological systems, industrial processes, and environmental science.	
pH	pH is a measure of the hydrogen ion concentration in a solution and determines its acidity or basicity: - Definition: $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}[\text{H}_3\text{O}^+]$. - Acidic Solutions: $\text{pH} < 7$, $[\text{H}^+] > [\text{OH}^-]$. Example: Lemon juice has $\text{pH} \approx 2$. - Neutral Solutions: $\text{pH} = 7$, $[\text{H}^+] = [\text{OH}^-]$. Example: Pure water. - Basic Solutions: $\text{pH} > 7$, $[\text{H}^+] < [\text{OH}^-]$. Example: Household ammonia has $\text{pH} \approx 11.5$. Importance: pH is critical in areas such as medicine (blood pH), agriculture (soil pH), and industry (chemical synthesis).	

Strong and Weak Acids and Bases

Strong acids and bases completely dissociate in aqueous solutions, while weak acids and bases only partially dissociate.

- **Strong Acids:** Examples include HCl, H₂SO₄, and HNO₃. Reaction: $\text{HCl} \rightarrow \text{H}^+ + \text{Cl}^-$.

- **Weak Acids:** Examples include CH₃COOH (acetic acid). Reaction: $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$.

- **Strong Bases:** Examples include NaOH and KOH. Reaction: $\text{NaOH} \rightarrow \text{Na}^+ + \text{OH}^-$.

- **Weak Bases:** Examples include NH₃. Reaction: $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$.

Applications: Understanding reactivity, buffer systems, and the strength of acids/bases in various environments.

Strengths of Conjugate Acid-Base Pairs			
acid		conjugate base	
HClO ₄	perchloric acid	ClO ₄ ⁻	perchlorate ion
HI	hydroiodic acid	I ⁻	iodide ion
HBr	hydrobromic acid	Br ⁻	bromide ion
HCl	hydrochloric acid	Cl ⁻	chloride ion
H ₂ SO ₄	sulfuric acid	HSO ₄ ⁻	hydrogen sulfate ion
HNO ₃	nitric acid	NO ₃ ⁻	nitrate ion
H ₃ O ⁺	Hydronium/ oxonium ion	H ₂ O	water
HSO ₄ ⁻	hydrogen sulfate ion	SO ₄ ²⁻	sulfate ion
HF	hydrofluoric acid	F ⁻	fluoride ion
HNO ₂	nitrous acid	NO ₂ ⁻	nitrite ion
HCOOH	formic acid	HCOO ⁻	formate ion
CH ₃ COOH	acetic acid	CH ₃ COO ⁻	acetate ion
NH ₄ ⁺	ammonium ion	NH ₃	ammonia
HCN	hydrocyanic acid	CN ⁻	cyanide ion
H ₂ O	water	OH ⁻	hydroxide ion
NH ₃	ammonia	NH ₂ ⁻	amide ion

Distinguish between weak and strong by measuring

1. **Electrical conductivity**
2. **Reacting acids with an active metal or metal carbonate, (*rate of reaction*)**
3. **Measuring pH**

STRONG		WEAK	
Full dissociation		Partial dissociation	
		Use equilibrium arrow/sign	
Acid	Base	Acids	Base
Good H^+ donors	Good H^+ acceptor	Poor H^+ donors	Poor H^+ acceptor
Has a weak conjugate base.	Has a weak conjugate acid.	Stronger conjugate base	Stronger conjugate acid
		Position of equilibrium lies to the	Position of equilibrium lies to the

- High electrical conductivity *because* $\uparrow$ [mobile ions]

- A faster rate of reaction *because* $\uparrow$ [H^+]

- Lower pH (acids)

- Higher pH (base)

left (favors reactants).

left (favors reactants).

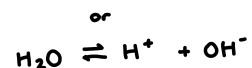
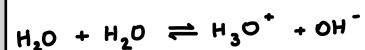
- Low electrical conductivity

- Slower rate of reaction

- Higher pH (acids)

- Lower pH (bases)

Ionization of water:



$$K_c = \frac{[H^+][OH^-]}{[H_2O]}$$

because the position of the equilibrium lies more to left; concentration of water is more or less constant

$$K_w = [H^+][OH^-]$$

The value of K_w at 298 K is $1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ (given in the data booklet)

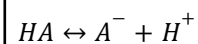
! like K_c , K_w is temperature dependent!

The relationship between H^+ and OH^- is **inverse**

	In aqueous solution $\uparrow [H^+]$; $\downarrow [OH^-]$		
	acidic	$pH < 7$	$[H^+] > [OH^-]$
	neutral	$pH = 7$	$[H^+] = [OH^-]$
	basic	$pH > 7$	$[H^+] < [OH^-]$
Remember: $\uparrow pH$; $\downarrow [H^+]$; $\uparrow [OH^-]$			
pKa and pKb	pKa and pKb provide measures of acid and base strength: - pKa: The negative logarithm of the acid dissociation constant (Ka). Lower pKa indicates a stronger acid. Example: For acetic acid ($K_a = 1.74 \times 10^{-5}$), $pK_a = -\log_{10}(1.74 \times 10^{-5}) \approx 4.76$. - pKb: The negative logarithm of the base dissociation constant (Kb). Lower pKb indicates a stronger base. - Relationship: $pK_a + pK_b = 14$ at 25°C. Applications: Used to design buffer systems and predict reaction equilibria in acid-base chemistry.	pKa: $HA + H_2O \leftrightarrow A^- + H_3O^+$ $K = \frac{[A^-][H_3O^+]}{[HA][H_2O]} K_a = K[H_2O] = \frac{[A^-][H_3O^+]}{[HA]}$	pKb: $B^- + H_2O \leftrightarrow HB + HO^-$ $K = \frac{[b][OH^-]}{[HB][H_2O]} K_b = K[H_2O] = \frac{[B][OH^-]}{[HB]}$
pH Calculations	pH calculations depend on the strength of the acid or base and their concentrations: - For strong acids/bases: Assume complete dissociation. $x = c_0$, $pH = -\log_{10}(c_0)$ Example: For 0.01 M HCl, $pH = -\log_{10}(0.01) = 2$. - For weak acids: $c_0 - x \approx c_0$ $K_a c_0 = x^2 \Rightarrow pH = -\log_{10} x = \frac{1}{2}(pK_a - \log_{10} c_0)$ Use the equilibrium expression $K_a = [H^+][A^-]/[HA]$. Example: If $K_a = 1.8 \times 10^{-5}$ and initial $[HA] = 0.1$ M, solve for $[H^+]$ to find pH. - For bases: Calculate pOH first, then use $pH = 14 - pOH$. Applications: Used in titrations, buffer preparation, and predicting solution properties.		

General Methods for calculating:

pH in a solution of acid with the stoichiometric concentration c_0 :

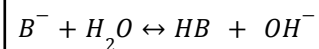


	HA	A-	H+
Stoichiometric	c_0	0	0
At equilibrium	$c_0 - x$	x	x

$$K_a = \frac{x^2}{c_0 - x}$$

$$pH = -\log_{10}\left(\frac{-K_a + \sqrt{K_a^2 + 4K_a c_0}}{2}\right)$$

pH in a solution of base with the stoichiometric concentration c_0 :



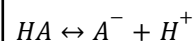
	B^-	HB	OH-
Stoichiometric	c_0	0	0
At equilibrium	$c_0 - x$	x	x

$$K_b = \frac{x^2}{c_0 - x}$$

$$pOH = -\log_{10}\left(\frac{-K_b + \sqrt{K_b^2 + 4K_b c_0}}{2}\right)$$

$$pH = 14 - pOH$$

pH in a solution of acid with the stoichiometric concentration c_0 with added A^- at concentration c_1 :



	HA	A-	H+
Stoichiometric	c_0	c_1	0
At equilibrium	$c_0 - x$	$c_1 + x$	x

$$K_a = \frac{(c_1 + x)x}{c_0 - x}$$

$$pH = -\log_{10}\left(\frac{-(K_a + c_1) + \sqrt{(K_a + c_1)^2 + 4K_a c_0}}{2}\right)$$

Shortcut:

$x \ll c_0, c_1$ (like a buffer sltn)

$$K_a = \frac{(c_1 + x)x}{c_0 - x} = \frac{c_1}{c_0}x \Rightarrow pH = pK_a + \log_{10} \frac{c_1}{c_0}$$

Solubility and Solubility Products

The solubility product constant (Ksp) represents the equilibrium between a solid and its dissolved ions:

- **Definition:** For salt $AB \rightleftharpoons A^+ + B^-$, $K_{sp} = [A^+][B^-]$.

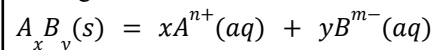
- **Example:** For $BaSO_4$ with $K_{sp} = 1.1 \times 10^{-10}$, $[Ba^{2+}] = [SO_4^{2-}] = x$. Solve $K_{sp} = x^2$ to find solubility $x = \sqrt{1.1 \times 10^{-10}} \approx 1.05 \times 10^{-5}$ M.

- **Applications:** Predicting precipitation, determining water hardness, and designing chemical separations.

Factors Affecting Solubility: Temperature, common ion effect, and pH.

Calculating Solubility

For a given reaction:



The solubility constant can be shown as:

$$K_{sp} = [A^{n+}(aq)]^x [B^{m-}(aq)]^y$$

	It is possible to use this equivalence to substitute into the formula above: $[A^{n+}] = \frac{x s_{A_x B_y}}{M_{s_{A_x B_y}}}, [B^{m-}] = \frac{y s_{A_x B_y}}{M_{s_{A_x B_y}}}$ $K_{sp} = \left[\frac{x s_{A_x B_y}}{M_{s_{A_x B_y}}} \right]^x \left[\frac{y s_{A_x B_y}}{M_{s_{A_x B_y}}} \right]^y$ $K_{sp} = x^x y^y \left[\frac{s_{A_x B_y}}{M_{s_{A_x B_y}}} \right]^{x+y}$
--	---

Equation	Variables/Units	Explanation/Use	Example
$K_w = [H^+][OH^-]$ $= 1.0 \times 10^{-14}$ Ion Product Constant of Water	Kw: Ion-product constant of water (unitless). [H ⁺]: Molar concentration of hydrogen ions (mol/L). [OH ⁻]: Molar concentration of hydroxide ions (mol/L).	At 25°C, the product of the concentrations of H ⁺ and OH ⁻ in pure water is constant at 1.0×10^{-14}	Example: If [H ⁺] = 1.0×10^{-5} M, calculate [OH ⁻]: $[OH^-] = \frac{K_w}{[H^+]} = \frac{1.0 \times 10^{-14}}{1.0 \times 10^{-5}} = 1.0 \times 10^{-9} \text{ M}$
Calculating [H ⁺] from pH: $pH = -\log[H^+]$ $[H^+] = 10^{-pH}$	[H ⁺]: Molar concentration of hydrogen ions (mol/L). pH: Dimensionless measure of acidity.	The pH scale quantifies the concentration of H ⁺ , with lower values being more acidic.	Example: If [H ⁺] = 1.0×10^{-3} M, calculate the pH: $pH = -\log(1.0 \times 10^{-3}) = 3.00$
Calculating [OH ⁻] from pOH: $pOH = -\log[OH^-]$ $[OH^-] = 10^{-pOH}$	[OH ⁻]: Molar concentration of hydroxide ions (mol/L). pOH: Dimensionless measure of basicity.		Example: If [OH ⁻] = 2.5×10^{-2} M, calculate the pOH: $pOH = -\log(2.5 \times 10^{-2}) = 1.60$

pH and pOH Relationship $pH + pOH = 14$		This equation links pH and pOH and is valid at 25°C.	Example: If $pH = 6.2$, calculate pOH: $pOH = 14.00 - pH = 14.00 - 6.2 = 7.8$
Percent Ionization: $\%ionization = \frac{[A]_{eq}}{[A]_0}$	$[A]_{eq}$: Ionized acid concentration at equilibrium (mol/L). $[A]_0$: Initial acid concentration (mol/L).		Example: For an acid with an initial concentration of 0.1 M and ionized concentration of 1.0×10^{-3} M: $\text{Percent ionization} = \frac{1.0 \times 10^{-3}}{0.1} \times 100 = 1.0\%$
Relationship between Acid Dissociation Constant and Base Dissociation Constant: $K_a * K_b = K_w$	K_a: Acid dissociation constant (unitless). K_b: Base dissociation constant (unitless). K_w: Ion-product constant of water $1.0 * 10^{-14}$		Example: For an acid with $K_a = 1.0 \times 10^{-5}$, calculate K_b: $K_b = \frac{K_w}{K_a} = \frac{1.0 \times 10^{-14}}{1.0 \times 10^{-5}} = 1.0 \times 10^{-9}$
Definition of the acid dissociation constant: $pK_a = -\log(K_a)$	K_a: Acid dissociation constant (unitless). pK_a: The logarithmic measure of the strength of an acid (dimensionless).	The pK_a is the negative logarithm of the acid dissociation constant K_a . A smaller pK_a indicates a stronger acid because it implies a larger K_a .	Example: If $K_a = 1.0 \times 10^{-5}$, calculate pK_a: $pK_a = -\log(1.0 \times 10^{-5}) = 5.00$
Henderson-Hasselbalch Equation: $pH = pK_a + \log\left(\frac{[con. B]}{[acid]}\right)$	pH: Dimensionless measure of acidity. pK_a: The negative logarithm of the acid dissociation constant (dimensionless). $[con. B]$: Molar concentration of the	This equation relates the pH of a solution to the pK_a of the acid and the ratio of the concentrations of the conjugate base and acid. It is used to calculate the pH of buffer solutions.	Example: If $pK_a = 4.74$, $[conjugate\ base] = 0.1\ M$, and $[acid] = 0.01\ M$, calculate the pH: $pH = 4.74 + \log\left(\frac{0.1}{0.01}\right)$ $pH = 4.74 + \log(10) = 4.74 + 1.00 = 5.74$

	conjugate base (mol/L).		
	[acid]: Molar concentration of the weak acid (mol/L).		

Lecture 10: Chapter 19 - Redox Reactions

Formula	Units	Notes	Example
Nernst Equation: $E = E^0 - \frac{RT}{nF} \ln Q$ Nernst Equation at 25 °C: $E = E^0 - \frac{0.0592}{nF} \ln Q$	E: Cell potential, in volts (V). E^0 : Standard cell potential, in volts (V). R: Gas constant, 8.314 J/(mol · K) T: Temperature, in kelvin (K). n: Number of electrons transferred, unitless. F: Faraday constant, 96485 C/mol Q: Reaction quotient, unitless.		Example: For a Cu-Zn cell with $E^0 = 1.10 \text{ V}$, $[Zn^{2+}]/[Cu^{2+}] = 10^6$: $E = 1.10 - \frac{0.0592}{2} \log(10^6) = 1.10 - 0.1776 = 0.92 \text{ V}$
Redox Reaction (Electrochemical Cells) $E_{cell} = E_{cathode} - E_{anode}$	E_{cell} - Cell potential (V) $E_{cathode}$ - Reduction potential at the cathode (V) E_{anode} - Reduction potential at the anode (V)		Example: For a galvanic cell with $E_{cathode} = 0.34 \text{ V}$ (Cu) and $E_{anode} = -0.76 \text{ V}$ (Zn): $E_{cell} = 0.34 - (-0.76) = 1.10 \text{ V}$

Activity Series & Electrode Potentials

Li $\rightarrow$ Li ⁺ + e ⁻	H ₂ evolves on contact with cold water
K $\rightarrow$ K ⁺ + e ⁻	
Ba $\rightarrow$ Ba ²⁺ + 2e ⁻	
Ca $\rightarrow$ Ca ²⁺ + 2e ⁻	
Na $\rightarrow$ Na ⁺ + e ⁻	
Mg $\rightarrow$ Mg ²⁺ + 2e ⁻	H ₂ evolves on contact with steam
Al $\rightarrow$ Al ³⁺ + 3e ⁻	
Zn $\rightarrow$ Zn ²⁺ + 2e ⁻	
Cr $\rightarrow$ Cr ³⁺ + 3e ⁻	
Fe $\rightarrow$ Fe ²⁺ + 2e ⁻	
Cd $\rightarrow$ Cd ²⁺ + 2e ⁻	H ₂ evolves on contact with acids
Co $\rightarrow$ Co ²⁺ + 2e ⁻	
Ni $\rightarrow$ Ni ²⁺ + 2e ⁻	
Sn $\rightarrow$ Sn ²⁺ + 2e ⁻	
Pb $\rightarrow$ Pb ²⁺ + 2e ⁻	
H ₂ $\rightarrow$ 2H ⁺ + 2e ⁻	no H ₂ generated on contact with water and acids
Cu $\rightarrow$ Cu ²⁺ + 2e ⁻	
Ag $\rightarrow$ Ag ⁺ + e ⁻	
Hg $\rightarrow$ Hg ²⁺ + 2e ⁻	
Pt $\rightarrow$ Pt ²⁺ + 2e ⁻	
Au $\rightarrow$ Au ³⁺ + 3e ⁻	

Standard Reduction Potentials at 25°C

Half-Reaction	E°/V	Half-Reaction	E°/V
F ₂ (g)+2e ⁻ $\rightarrow$ 2F ⁻ (aq)	+2.87	O ₃ (g)+2H ⁺ (aq)+2e ⁻ $\rightarrow$ O ₂ (g)+H ₂ O	+2.07
Co ³⁺ (aq)+e ⁻ $\rightarrow$ Co ²⁺ (aq)	+1.82	H ₂ O ₂ (aq)+2H ⁺ (aq)+2e ⁻ $\rightarrow$ 2H ₂ O	+1.77
PbO ₂ (s)+4H ⁺ (aq)+SO ₄ ²⁻ (aq)+2e ⁻ $\rightarrow$ PbSO ₄ (s)+2H ₂ O	+1.70	Ce ⁴⁺ (aq)+e ⁻ $\rightarrow$ Ce ³⁺ (aq)	+1.61
MnO ₄ ⁻ (aq)+8H ⁺ (aq)+5e ⁻ $\rightarrow$ Mn ²⁺ (aq)+4H ₂ O	+1.51	Au ³⁺ (aq)+3e ⁻ $\rightarrow$ Au(s)	+1.50
Cl ₂ (g)+2e ⁻ $\rightarrow$ 2Cl ⁻ (aq)	+1.36	Cr ₂ O ₇ ²⁻ (aq)+14H ⁺ (aq)+6e ⁻ $\rightarrow$ 2Cr ³⁺ (aq)+7H ₂ O	+1.33
MnO ₂ (s)+4H ⁺ (aq)+2e ⁻ $\rightarrow$ Mn ²⁺ (aq)+2H ₂ O	+1.23	O ₂ (g)+4H ⁺ (aq)+4e ⁻ $\rightarrow$ 2H ₂ O	+1.23
Br ₂ (l)+2e ⁻ $\rightarrow$ 2Br ⁻ (aq)	+1.07	NO ₃ ⁻ (aq)+4H ⁺ (aq)+3e ⁻ $\rightarrow$ NO(g)+2H ₂ O	+0.96
2Hg ²⁺ (aq)+2e ⁻ $\rightarrow$ Hg ₂ ²⁺ (aq)	+0.92	Hg ₂ ²⁺ (aq)+2e ⁻ $\rightarrow$ 2Hg(l)	+0.85
Ag ⁺ (aq)+e ⁻ $\rightarrow$ Ag(s)	+0.80	Fe ³⁺ (aq)+e ⁻ $\rightarrow$ Fe ²⁺ (aq)	+0.77
O ₂ (g)+2H ⁺ (aq)+2e ⁻ $\rightarrow$ H ₂ O ₂ (aq)	+0.68	MnO ₄ ⁻ (aq)+2H ₂ O+3e ⁻ $\rightarrow$ MnO ₂ (s)+4OH ⁻ (aq)	+0.59
I ₂ (s)+2e ⁻ $\rightarrow$ 2I ⁻ (aq)	+0.53	O ₂ (g)+2H ₂ O+4e ⁻ $\rightarrow$ 4OH ⁻ (aq)	+0.40
Cu ²⁺ (aq)+2e ⁻ $\rightarrow$ Cu(s)	+0.34	AgCl(s)+e ⁻ $\rightarrow$ Ag(s)+Cl ⁻ (aq)	+0.22
SO ₄ ²⁻ (aq)+4H ⁺ (aq)+2e ⁻ $\rightarrow$ SO ₂ (g)+2H ₂ O	+0.20	Cu ²⁺ (aq)+e ⁻ $\rightarrow$ Cu ⁺ (aq)	+0.15
Sn ⁴⁺ (aq)+2e ⁻ $\rightarrow$ Sn ²⁺ (aq)	+0.13	2H ⁺ (aq)+2e ⁻ $\rightarrow$ H ₂ (g)	0.00
Pb ²⁺ (aq)+2e ⁻ $\rightarrow$ Pb(s)	-0.13	Sn ²⁺ (aq)+2e ⁻ $\rightarrow$ Sn(s)	-0.14
Ni ²⁺ (aq)+2e ⁻ $\rightarrow$ Ni(s)	-0.25	Co ²⁺ (aq)+2e ⁻ $\rightarrow$ Co(s)	-0.28
PbSO ₄ (s)+2e ⁻ $\rightarrow$ Pb(s)+ SO ₄ ²⁻ (aq)	-0.31	Cd ²⁺ (aq)+2e ⁻ $\rightarrow$ Cd(s)	-0.40
Fe ²⁺ (aq)+2e ⁻ $\rightarrow$ Fe(s)	-0.44	Cr ³⁺ (aq)+3e ⁻ $\rightarrow$ Cr(s)	-0.74
Zn ²⁺ (aq)+2e ⁻ $\rightarrow$ Zn(s)	-0.76	2H ₂ O+2e ⁻ $\rightarrow$ H ₂ (g)+2OH ⁻ (aq)	-0.83
Mn ²⁺ (aq)+2e ⁻ $\rightarrow$ Mn(s)	-1.18	Al ³⁺ (aq)+3e ⁻ $\rightarrow$ Al(s)	-1.66
Be ²⁺ (aq)+ 2e ⁻ $\rightarrow$ Be(s)	-1.85	Mg ²⁺ (aq)+2e ⁻ $\rightarrow$ Mg(s)	-2.37
Na ⁺ (aq) +e ⁻ $\rightarrow$ Na(s)	-2.71	Ca ²⁺ (aq)+2e ⁻ $\rightarrow$ Ca(s)	-2.87
Sr ²⁺ (aq) +2e ⁻ $\rightarrow$ Sr(s)	-2.89	Ba ²⁺ (aq)+2e ⁻ $\rightarrow$ Ba(s)	-2.90
K ⁺ (aq) +e ⁻ $\rightarrow$ K(s)	-2.93	Li ⁺ (aq)+e ⁻ $\rightarrow$ Li(s)	-3.05

Balancing other atoms in redox reactions	Depending if you want to balance an acidic or a basic solution. Follow the steps for Balancing and Solving redox reactions! 1. Assign oxidation states to each of the species on both sides 2. Balance the redox atoms, so there are the same amount. 3. Identify how many electrons transferred and identify whether it is an oxidation or a reduction 4. Balance the electrons on both sides, so that the electrons that are gained and lost are the same. 5. Balance other atoms.	Applying oxidation states for balancing $4\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 6\text{ClO}_2^{-}(\text{aq}) \rightarrow 8\text{Cr}^{3+}(\text{aq}) + 3\text{Cl}_2\text{O}_7(\text{aq})$ acidic solution: <table border="1" style="display: inline-table; margin-right: 10px;"> <tr> <td>to balance</td><td>O atoms</td><td>H atoms</td></tr> <tr> <td>use</td><td>H₂O</td><td>2H⁺</td></tr> </table> Make sure all atoms and charges balance! $38\text{H}^{+}(\text{aq}) + 4\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 6\text{ClO}_2^{-}(\text{aq}) \rightarrow 8\text{Cr}^{3+}(\text{aq}) + 3\text{Cl}_2\text{O}_7(\text{aq}) + 19\text{H}_2\text{O}(\text{aq})$ basic solution: <table border="1" style="display: inline-table; margin-right: 10px;"> <tr> <td>to balance</td><td>O atoms</td><td>H atoms</td></tr> <tr> <td>use</td><td>2OH⁻</td><td>H₂O</td></tr> </table> $19\text{H}_2\text{O}(\text{aq}) + 4\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 6\text{ClO}_2^{-}(\text{aq}) \rightarrow 8\text{Cr}^{3+}(\text{aq}) + 3\text{Cl}_2\text{O}_7(\text{aq}) + 38\text{OH}^{-}(\text{aq})$ The oxidation state method: ✓ 1. Assign oxidation states to each of the species. ✓ 2. Balance the redox atoms. ✓ 3. Identify the electrons transferred. ✓ 4. Balance the electrons. ✓ 5. Balance all other atoms.	to balance	O atoms	H atoms	use	H ₂ O	2H ⁺	to balance	O atoms	H atoms	use	2OH ⁻	H ₂ O
to balance	O atoms	H atoms												
use	H ₂ O	2H ⁺												
to balance	O atoms	H atoms												
use	2OH ⁻	H ₂ O												
Balancing redox reactions using the Half/time formula...														

Strong and Weak, Acids and Bases:

Acids and Bases



Acid				Base		
Strength of Acid	Ka	Name	Formula	Formula	Name	Strength of Base
Strongest	Large	Perchloric acid	HClO ₄	ClO ₄ ⁻	Perchlorate ion	Weakest
	3.2×10 ⁹	Hydroiodic acid		I ⁻	Iodide	
	1.0×10 ⁹	Hydrobromic acid	HBr	Br ⁻	Bromide	
	1.3×10 ⁶	Hydrochloric acid	HCl	Cl ⁻	Chloride	
	1.0×10 ³	Sulfuric acid	H ₂ SO ₄	HSO ₄ ⁻	Hydrogen sulfate ion	
	2.4×10 ¹	Nitric acid	HNO ₃	NO ₃ ⁻	Nitrate ion	
	-----	Hydronium ion	H ₃ O ⁺	H ₂ O	Water	
	5.4×10 ⁻²	Oxalic acid	HO ₂ C ₂ O ₂ H	HO ₂ C ₂ O ₂ ⁻	Hydrogen oxalate ion	
	1.3×10 ⁻²	Sulfurous acid	H ₂ SO ₃	HSO ₃ ⁻	Hydrogen sulfite ion	
	1.0×10 ⁻²	Hydrogen sulfate ion	HSO ₄ ⁻	SO ₄ ²⁻	Sulfate ion	
	7.1×10 ⁻³	Phosphoric acid	H ₃ PO ₄	H ₂ PO ₄ ⁻	Dihydrogen phosphate ion	
	7.2×10 ⁻⁴	Nitrous acid	HNO ₂	NO ₂ ⁻	Nitrite ion	
	6.6×10 ⁻⁴	Hydrofluoric acid	HF	F ⁻	Fluoride ion	
	1.8×10 ⁻⁴	Methanoic acid	HCO ₂ H	HCO ₂ ⁻	Methanoate ion	
	6.3×10 ⁻⁵	Benzoic acid	C ₆ H ₅ COOH	C ₆ H ₅ COO ⁻	Benzoate ion	
	5.4×10 ⁻⁵	Hydrogen oxalate ion	HO ₂ C ₂ O ₂ ⁻	O ₂ C ₂ O ₂ ²⁻	Oxalate ion	
	1.8×10 ⁻⁵	Ethanoic acid	CH ₃ COOH	CH ₃ COO ⁻	Ethanoate (acetate) ion	
	4.4×10 ⁻⁷	Carbonic acid	H ₂ CO ₃	HCO ₃ ⁻	Hydrogen carbonate ion	
	1.1×10 ⁻⁷	Hydrosulfuric acid	H ₂ S	HS ⁻	Hydrogen sulfide ion	
	6.3×10 ⁻⁸	Dihydrogen phosphate ion	H ₂ PO ₄ ⁻	HPO ₄ ²⁻	Hydrogen phosphate ion	
	6.2×10 ⁻⁸	Hydrogen sulfite ion	HS ⁻	SO ₃ ²⁻	Sulfite ion	
	2.9×10 ⁻⁸	Hypochlorous acid	HClO	ClO ⁻	Hypochlorite ion	
	6.2×10 ⁻¹⁰	Hydrocyanic acid	HCN	CN ⁻	Cyanide ion	
	5.8×10 ⁻¹⁰	Ammonium ion	NH ₄ ⁺	NH ₃	Ammonia	
	5.8×10 ⁻¹⁰	Boric acid	H ₃ BO ₃	H ₂ BO ₃ ⁻	Dihydrogen borate	
	4.7×10 ⁻¹¹	Hydrogen carbonate ion	HCO ₃ ⁻	CO ₃ ²⁻	Carbonate ion	
	4.2×10 ⁻¹³	Hydrogen phosphate ion	HPO ₄ ²⁻	PO ₄ ³⁻	Phosphate ion	
	1.8×10 ⁻¹³	Dihydrogen borate ion	H ₂ BO ₃ ⁻	HBO ₃ ²⁻	Hydrogen borate ion	
	1.3×10 ⁻¹³	Hydrogen sulfide ion	HS ⁻	S ²⁻	Sulfide ion	
	1.6×10 ⁻¹⁴	Hydrogen borate ion	HBO ₃ ²⁻	BO ₃ ³⁻	Borate ion	
Weakest	-----	water	H ₂ O	OH ⁻	Hydroxide	Strongest



Acid			Base		
Increasing acid strength ↑	perchloric acid	HClO_4	Do not undergo base ionization in water	ClO_4^-	perchlorate ion
	sulfuric acid	H_2SO_4		HSO_4^-	hydrogen sulfate ion
	hydrogen iodide	HI		I^-	iodide ion
	hydrogen bromide	HBr		Br^-	bromide ion
	hydrogen chloride	HCl		Cl^-	chloride ion
	nitric acid	HNO_3	Undergo complete acid ionization in water	NO_3^-	nitrate ion
	hydronium ion	H_3O^+		H_2O	water
	hydrogen sulfate ion	HSO_4^-		SO_4^{2-}	sulfate ion
	phosphoric acid	H_3PO_4		H_2PO_4^-	dihydrogen phosphate ion
	hydrogen fluoride	HF		F^-	fluoride ion
	nitrous acid	HNO_2		NO_2^-	nitrite ion
	acetic acid	$\text{CH}_3\text{CO}_2\text{H}$		CH_3CO_2^-	acetate ion
	carbonic acid	H_2CO_3		HCO_3^-	hydrogen carbonate ion
	hydrogen sulfide	H_2S		HS^-	hydrogen sulfide ion
	ammonium ion	NH_4^+		HN_3	ammonia
	hydrogen cyanide	HCN		CN^-	cyanide ion
	hydrogen carbonate ion	HCO_3^-		CO_3^{2-}	carbonate ion
	water	H_2O		OH^-	hydroxide ion
	hydrogen sulfide ion	HS^-	Undergo complete base ionization in water	S^{2-}	sulfide ion
	ethanol	$\text{C}_2\text{H}_5\text{OH}$		$\text{C}_2\text{H}_5\text{O}^-$	ethoxide ion
	ammonia	NH_3		NH_2^-	amide ion
	hydrogen	H_2		H^-	hydride ion
	methane	CH_4		CH_3^-	methide ion
			Increasing base strength ↓		

Acid		Conjugate base		
Strong acids	Perchloric acid	HClO ₄	ClO ₄ ⁻	Perchlorate ion
	Sulfuric acid	H ₂ SO ₄	HSO ₄ ⁻	Hydrogen sulfate ion
	Hydroiodic acid	HI	I ⁻	Iodide ion
	Hydrobromic acid	HBr	Br ⁻	Bromide ion
	Hydrochloric acid	HCl	Cl ⁻	Chloride ion
	Nitric acid	HNO ₃	NO ₃ ⁻	Nitrate ion
	Hydronium ion	H ₃ O ⁺	H ₂ O	Water
Weak acids	Hydrosulfate	HSO ₄ ⁻	SO ₄ ²⁻	Sulfate ion
	Phosphoric acid	H ₃ PO ₄	H ₂ PO ₄ ⁻	Dihydrogen phosphate ion
	Hydrofluoric acid	HF	F ⁻	Fluoride ion
	Nitrous acid	HNO ₂	NO ₂ ⁻	Nitrite ion
	Acetic acid	CH ₃ COOH	CH ₃ COO ⁻	Acetate ion
	Carbonic acid	H ₂ CO ₃	HCO ₃ ⁻	Hydrogen carbonate ion
	Hydrogen sulfide	H ₂ S	HS ⁻	Hydrogen sulfide ion
	Ammonium ion	NH ₄ ⁺	NH ₃	Ammonia
	Hydrogen cyanide	HCN	CN ⁻	Cyanide ion
	Hydrogen carbonate	HCO ₃ ⁻	CO ₃ ²⁻	Carbonate ion
	Water	H ₂ O	HO ⁻	Hydroxide ion
Very weak acid	Hydrogen sulfide ion	HS ⁻	S ²⁻	Sulfide ion
	Ethanol	CH ₃ CH ₂ -OH	CH ₃ CH ₂ -O ⁻	Ethoxide ion
	Ammonia	NH ₃	NH ₂ ⁻	Amide ion
	Hydrogen	H ₂	H ⁻	Hydride ion
	Methane	CH ₄	CH ₃ ⁻	Methide ion

Extra:

Coordinate Numbers:

Step 1: Identify the central atom or ion (usually a metal).

Step 2: Look at the ligands attached to the central atom. Ligands are molecules or ions that donate a pair of electrons to form a bond with the metal.

Step 3: Count the total number of direct bonds formed between the central atom and the ligands.

Summary of Acids and Bases:

Acids → Any substance dissolved in water, which increases the concentration of H^+ in a solution.

The simplest form of hydrated proton (H_3O^+)

- When hydrogen is bound to a very electronegative element (acid), when it is dissolved in water it dissociates into X^- and H^+ ion.
- On a molecular level: H^+ is solvated to H_3O^+ molecules.

Acetic Acid: (weak acid)

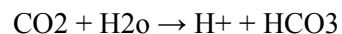
- Only partially dissociates.
- The degree to which it ionises is less than that of a stronger acid.

Base → Any substance that raises the concentration of hydroxide ion $[OH^-]$.

- Neutralises acid.
- Strong base completely dissociates in water.

Ammonia (NH_3): (weak base)

- Acts as a base in water → to create Ammonium ion NH_4^+ (the conjugate acid)



$CO_2 \rightarrow$ weak acid

Lewis acid do not necessarily contain any H^+ in itself, but still increase the concentration of H^+ when placed in water)

- A base is a substance that can donate a pair of electrons, and (All Bronsted bases are Lewis bases).
- An acid is a substance that can accept a pair of electrons.

A Lewis acid-base reaction, therefore, is one that involves the donation of a pair of electrons from one species to another. Such a reaction does not produce salt and water.

Lewis' concept includes as acid-base reactions many reactions that do not involve Bronsted acids.

→ Green neutral, Purple basic, red acidic. (yellow weak acid)

Buffer Systems:

- Buffers resist changes in pH upon the addition of small amounts of acid or base.
- Composed of a weak acid and its conjugate base or a weak base and its conjugate acid.

The strength of an acid is influenced by its molecular structure and the stability of its conjugate base.

Hydrohalic Acids (HX):

- Acid strength increases as the bond strength between hydrogen and the halogen decreases. For example, HF is weaker than HI because the H-I bond is weaker and dissociates more easily.

Oxoacids (H_xEO_y):

- Acid strength depends on the electronegativity of the central atom (E) and the number of oxygen atoms bonded to it.
- Higher electronegativity and a greater number of oxygen atoms enhance the acid strength due to increased electron withdrawal from the hydrogen.

Salts That Produce Neutral Solutions:

- Salts derived from strong acids and strong bases (e.g., NaCl, KNO_3) do not hydrolyze and result in neutral solutions.

Salts That Produce Basic Solutions:

- Salts from strong bases and weak acids (e.g., $\text{NaC}_2\text{H}_3\text{O}_2$, KCN) hydrolyze, generating OH^- ions, making the solution basic.

Salts That Produce Acidic Solutions:

- Salts from strong acids and weak bases (e.g., NH_4Cl) hydrolyze to produce H^+ ions, creating an acidic solution.

Metal Ion Hydrolysis:

- Some metal ions (e.g., Fe^{3+} , Al^{3+}) hydrolyze water to release H^+ ions, leading to acidic solutions.

Salts in Which Both the Cation and Anion Hydrolyze:

- The pH of the solution depends on the relative strengths of the acidic cation and basic anion.

When we say that an atom or molecule “ionizes more easily,” it means that it requires less energy to lose or gain electrons to form ions. The ease of ionization depends on the context, but it typically refers to either the dissociation of a molecule in water or the removal of an electron from an atom.

Hydrolysis is a chemical reaction in which a molecule reacts with water, resulting in the breaking of a bond within the molecule.

How to Determine the Effect of a Salt:

1. Identify the acid and base that formed the salt.
2. Decide if they are strong or weak:
 - Strong acid + Strong base = Neutral salt.
 - Weak acid + Strong base = Basic salt.

- Strong acid + Weak base = Acidic salt.

Cations (positive ions) hydrolyze if they come from a weak base.

Anions (negative ions) hydrolyze if they come from a weak acid.

Strong acids and strong bases do NOT hydrolyze.

- **Weak acids' anions** hydrolyze to produce OH^- (basic solution).
- **Weak bases' cations** hydrolyze to produce H^+ (acidic solution).
- For salts where both ions hydrolyze, compare the acid and base strength using their K_a and K_b values:
 - If $K_a > K_b$: Solution is acidic.
 - If $K_b < K_a$: Solution is basic.